THE BIT DELIVERED TWICE: TWO KINDS AND NO THIRD

STEP 10 · THE BIT DELIVERED TWICE, BY ITS TWO KINDS, AND NO THIRD

KIND 1 · THE FORMAL CARGO

The record's value carried to the host's frame. It arrives free, the antecedent a theorem; it gives rank equal to order in one step; and it is exactly the sentence.

`FormalCargo` · `cargo_arrives` · `formal_kind_delivers`
`formal_iff_sentence`

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KIND 2 · THE PHYSICAL WITNESS

Each curve's group of rational points is a physical register whose free directions are each registered at one Landauer floor quantum, the curve at one more, so its heat is rank plus one; the carrier registers the free directions as the zero modes at the centre. Its last field is the bit, carried as a field, never proved.

`PhysicalWitness` · `witness_closes` · `physical_kind_delivers`
`physical_iff_sentence`

The two kinds are one bit
The physical witness is a formal cargo, and the cargo is exactly the sentence.
`the_two_kinds_are_one_bit`

Two kinds and no third
Every witness of the sentence, in any vocabulary, is a delivery of both kinds.
`two_kinds_and_no_third`
`every_route_is_both_kinds`

The one refuter
One curve whose rank is not its order refutes both kinds at once, and the sentence itself.
`the_one_refuter` · `refuter_denies_the_sentence`

THE WITNESSES OF THE SUBJECT, TYPED · EACH STRICTLY INSIDE THE BIT OR STRICTLY STRONGER

The low-order witness
Inside the bit and strictly smaller: a curve of rank zero at order two carries it and fails the sentence.
`low_is_strictly_inside` · `low_witness_strict`

The retired premise Ω
It gives the value and is strictly stronger; it is retired, never the ground.
`omega_gives_value` · `omega_strictly_stronger`

The full act
Rank and leading coefficient: strictly stronger, carried beside, and the act does not give it.
`full_act_strictly_stronger`
`act_does_not_give_full_value`
`the_witnesses_typed`

READING theorem on no axiom at all · binding theorem on Lean's standard axioms · definition or structure — names shown by their last component